

Coding Practice with Advanced Shapes

An **advanced shape** in robotics has:

- More sides than basic shapes like squares or triangles
 - Unusual turn angles
 - Requires loops, functions, and precise movement
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Geometry Refresher

Shape	Number of Sides	Turn Angle Between Sides
Square	4	90°
Hexagon	6	60° turns (external)
Octagon	8	45° turns (external)
Star	10 movements (5 points)	~144° turns

Remember: Turn angle = $360^\circ \div \text{number of sides}$.

Example

- A hexagon has 6 equal sides.
- The robot must turn 60° after each forward move to make the shape.

Coding Strategy:

1. Move the robot forward a set distance and then turn 60°.
2. Use a repeat block (loop)
3. Test and adjust. Make sure the robot returns back to the starting point.

Exercise: Draw an Octagon

What You Know:

- Octagon = 8 sides
- Each external turn = 45°

Your Mission:

- Make the robot trace the outline of an octagon using a loop block.
 - Test the robot.
 - Adjust measurements to make the shape clean and closed.
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Challenge: Draw a Star

What You Know:

- Star = 5 points
- Each external turn $\approx 144^\circ$

Your Mission:

- Make the robot trace the outline of a 5-pointed star
- Test the robot.
- Adjust measurements to make the shape clean and closed.

Submit a screenshot of your code to the corresponding Google Classroom assignment.

Checklist:

- ✓ Robot returns to or near starting point
- ✓ Shape is consistent on multiple tries
- ✓ Code is clean (uses functions/loops, no repeated blocks)
- ✓ Notebook includes a screenshot of your code

